

Tail-Latency-Aware Segmented Waveguide-Enabled Pinching-Antenna Systems for Hyper-Reliable Low-Latency Communications Under Bursty Traffic

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Abstract—Segmented waveguide-enabled pinching-antenna systems (SWANs) replace the single long dielectric waveguide of a pinching-antenna system (PASS) by several short, individually fed segments, each of which activates one pinching antenna (PA). SWANs thereby provide reconfigurable line-of-sight channels without inter-antenna radiation and can serve users either by aggregating all segments or by multiplexing them. Existing SWAN and PASS designs, however, optimize average rates under full-buffer traffic, whereas 6G hyper-reliable low-latency communication (HRLLC) is specified by the *tail* of the packet delay under bursty, deadline-constrained arrivals. This paper studies a downlink multi-user SWAN whose users generate packets according to Markov-modulated ON-OFF processes, transmit with finite-blocklength (FBL) codes, and discard packets whose delay budget expires. We formulate the joint PA placement, segment-mode selection, user scheduling, and rate/power adaptation problem under per-user delay-violation-probability constraints and propose a two-timescale tail-latency-aware scheme, TLA-SWAN. On the frame timescale, a closed-form mapping from each user's burst statistics and tail target to a *tail-latency-aware required service rate* drives a low-complexity block-coordinate PA placement that maximizes the minimum tail-latency margin. On the slot timescale, a packet-value scheduler assigns to every queued packet a value that grows exponentially with its age at a rate set by the user's tail exponent, and minimizes the drift of the resulting tail-risk potential plus a power penalty by jointly selecting the served user set (aggregation or multiplexing), the number of packets per codeword, and the power level through an exhaustive search over pre-computed zero-forcing tables. Extensive slot-level simulations with FBL decoding errors and deadline-constrained queues show that TLA-SWAN reduces the delay-violation probability by more than an order of magnitude with respect to rate-centric SWAN designs, classical deadline-aware schedulers, single-waveguide PASS, and fixed antenna arrays, that its statistical placement is robust to reconfiguration delays that cripple reactive repositioning, and that it exposes a tunable power-tail trade-off.

Index Terms—Pinching-antenna systems, segmented waveguide, hyper-reliable low-latency communication, tail latency, bursty traffic, finite blocklength, Lyapunov optimization, scheduling.

I. INTRODUCTION

HYPER-RELIABLE and low-latency communication (HRLLC) is one of the six usage scenarios of IMT-2030, with research targets of 0.1–1 ms air-interface latency and reliability between $1 - 10^{-5}$ and $1 - 10^{-7}$ [1]. Such targets

are not statements about the *average* delay: they bound the probability that the delay of a packet exceeds a budget, i.e., the tail of the delay distribution [2]. Two features of HRLLC applications make the tail particularly hard to control. First, the traffic is bursty: industrial closed-loop control, event-triggered alarms, and extended-reality (XR) frames produce packets in clusters separated by silent periods [3]–[5], so that even a lightly loaded link experiences short overload periods during which queues build up. Second, HRLLC packets are short, so that finite-blocklength (FBL) coding penalties and decoding errors must be accounted for jointly with queueing [6]–[9]. The wireless link therefore needs enough *instantaneous* capacity to drain a burst within the delay budget, which in turn calls for physical-layer architectures that can bring high signal-to-noise ratio (SNR) to wherever a burst occurs.

Pinching-antenna systems (PASS) are a recently proposed flexible-antenna technology in which small dielectric particles (pinching antennas, PAs) are attached at adjustable positions along a dielectric waveguide fed by the base station (BS) [10]–[13]. By activating a PA close to a user, PASS shortens the free-space path and establishes a strong line-of-sight (LoS) link, which makes it attractive for millimeter-wave deployments in factories and indoor venues. A single long waveguide, however, suffers from in-waveguide propagation loss and from inter-antenna radiation (IAR) when several PAs are activated. Segmented waveguide-enabled PASS (SWAN) [14] solves both problems by dividing the waveguide into short segments, each with its own feed connected to the BS by fiber or cable, and by activating a single PA per segment. The feeds can be driven by separate radio-frequency (RF) chains, so that a SWAN behaves like a distributed antenna array whose element positions are reconfigurable within their segments. This gives rise to three operating modes [14], [15]: segment selection (one segment serves a user), segment aggregation (SA, all segments coherently serve one user), and segment multiplexing (SM, the segments spatially multiplex several users). Existing SWAN studies characterize the SNR and sum-rate of these modes [14], [16], [17], design hybrid beamformers [18], [19], integrate sensing [20]–[22], and design random-access protocols [15]. All of them assume saturated (full-buffer) traffic and infinite-blocklength coding; none considers queueing, deadlines, or the delay tail.

The combination of SWAN and HRLLC raises three questions that the rate-centric literature does not answer. (i) *Where should the PAs be placed* when users differ in burstiness and in tail targets? A placement that maximizes the sum-rate moves

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PAs towards users with good channels, whereas the delay tail is governed by the users whose bursts are hardest to drain. (ii) *How should the BS decide, slot by slot, between aggregation and multiplexing*, how many short packets to bundle in a codeword, and at which power, when a decoding failure delays all bundled packets and the reliability target is 10^{-5} ? (iii) *How fast can the SWAN be reconfigured?* Physical PA repositioning takes from microseconds (pre-installed PAs that are switched on and off) to seconds (sliding PAs) [11], [23]–[25], so that chasing bursts by moving PAs may cost more than it gains.

A. Related Work

1) *PASS and SWAN*: The channel and signal models of PASS were established in [10] and are surveyed in [11]–[13]. SWANs were introduced in [14], which derived the SNR scaling laws of the segment-selection, aggregation, and multiplexing modes and showed that segmentation shortens both the PA-to-user and the PA-to-feed distances. Follow-up work addressed uplink sum-rate maximization with TDMA and NOMA [16], joint segment activation and PA placement under a single RF chain [17], tri-hybrid (digital, analog, pinching) beamforming [18], learning-based placement [19], integrated sensing and communication [20]–[22], maintainability [26], frequency selectivity over long apertures [27], and access protocols that exploit the reconfigurable channel diversity of SWANs [15]. Timeliness has been considered only through the age of information for vehicular platooning over SWANs [28]. For single-waveguide PASS, FBL rate optimization for a single user was studied in [29] and a statistical-CSI short-packet design in [30]; both consider one user, one packet, and no queue. The cost of PA repositioning has been modeled through a finite movement speed [23], [24] and discrete activation [25], again without arrivals or deadlines. A recent network-level perspective [31] lists queue-aware PA scheduling under discrete activation constraints as an open problem, which is precisely the problem addressed here.

2) *Tail latency under bursty traffic and finite blocklength*: The delay tail of a wireless queue is classically characterized by effective bandwidth and effective capacity [32]–[35], by stochastic network calculus [36]–[38], and by extreme value theory [2], [39]. Markov-modulated ON–OFF sources are the standard model of bursty arrivals, with closed-form effective bandwidths due to Anick, Mitra, and Sondhi [40], [41]. In the URLLC context, [42] converted a queueing-delay-violation constraint into a service-rate requirement through the effective bandwidth of the arrivals and combined it with the FBL decoding error; [38], [43] incorporated FBL coding into effective capacity and network calculus; [8] adapted the blocklength to the queue state; and [9], [44] characterized the reliability–latency–rate trade-off and tractable queue-tail approximations. Tail-aware wireless schedulers include the modified largest-weighted-delay-first (M-LWDF) rule [45], whose weights are chosen from the per-user delay budget and violation target, the exponential rule [46], and largest-weighted-delay-first scheduling, which is large-deviations optimal [47]. Lyapunov drift-plus-penalty methods [48], [49] turn time-average or probabilistic delay constraints into virtual queues and have

been applied to URLLC edge computing [39], delay-sensitive scheduling [50], [51], and hard-latency-constrained scheduling [5]. None of these works involves a reconfigurable antenna geometry.

3) *Reconfigurable antennas for URLLC*: Fluid antenna systems have been studied for HRLLC with FBL coding and an explicit port-switching delay that shortens the effective blocklength [52], [53], movable antennas for short-packet NOMA [54], and reconfigurable intelligent surfaces for URLLC with two-timescale designs in which slow reconfiguration follows statistical CSI while fast precoding follows instantaneous CSI [55], [56]. These works confirm that reconfiguration overhead must be charged against the latency budget, but they do not model queues or bursty arrivals, and they do not exploit the aggregation/multiplexing duality that is specific to SWANs.

B. Contributions

To the best of our knowledge, this is the first work that studies a SWAN (or any PASS) with packet arrivals, queues, deadlines, and FBL coding, and the first that designs the PA geometry from the delay tail rather than from the rate. The contributions are as follows.

- *Model and formulation*. We model a downlink multi-user SWAN with per-segment RF chains and power amplifiers, LoS-dominant PA channels, Markov-modulated ON–OFF packet arrivals, FBL transmissions with retransmission upon decoding failure, deadline-based packet discarding, and a reconfiguration delay per PA move. We formulate the joint PA placement, mode selection, scheduling, and rate/power adaptation problem under per-user delay-violation-probability constraints.
- *Tail-latency-aware required rate*. We map the ON–OFF burst parameters, the delay budget $D_{\max}$, and the violation target δ of a user into a single scalar, the tail-latency-aware required service rate c_{req} , in closed form. This scalar summarizes “how fast a burst must be drained”; the tail exponent $\Lambda_k = \ln(1/\delta_k)/D_{\max}$ that defines it is the common currency of both timescales of the proposed scheme.
- *Two-timescale TLA-SWAN scheme*. On the frame timescale, a block-coordinate-descent placement (TAPP) maximizes the minimum ratio between the aggregation-mode FBL rate of a user and its required rate, which moves PAs towards bursty and stringent users rather than towards high-rate users. On the slot timescale, a tail-aware scheduler (TAS) values every queued packet exponentially in its age with the user’s tail exponent, so that the sum of the values is a tail-risk potential whose growth measures the imminence of deadline violations, and minimizes the drift of this potential plus a power penalty by jointly selecting the user subset (aggregation or multiplexing), the packets per codeword, and the power level through an exhaustive search over zero-forcing tables that are computed once per frame. The per-slot complexity is that of a table lookup.
- *Performance guarantees*. We state the structural properties of the scheme (monotone convergence of TAPP, closed-form single-user placement, the drift-plus-penalty characterization of TAS with the resulting bound on the tail-risk potential,

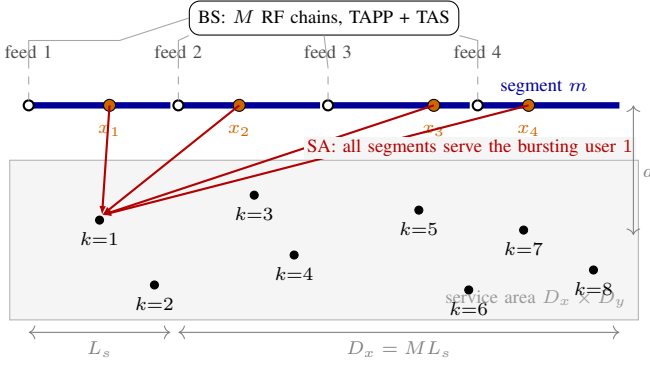


Fig. 1: Downlink SWAN with $M = 4$ segments. Each segment is fed at its left end by its own RF chain and activates one PA at x_m ; the BS chooses per slot between segment aggregation (SA, shown) and segment multiplexing (SM).

and an effective-capacity-based bound on the delay-violation probability); the proofs are provided in the appendices.

- *Evaluation.* A slot-level simulator with FBL decoding errors, deadline dropping, and reconfiguration delays is used to compare TLA-SWAN with rate-centric SWAN placement, max-weight, earliest-deadline-first, and M-LWDF scheduling, reactive PA repositioning, single-waveguide PASS, and a fixed antenna array, over the peak rate, burst duration, transmit power, number of segments and users, delay budget, reconfiguration delay, fading, traffic model, and parameter mismatch.

C. Organization and Notation

Section II presents the system model, Section III the problem formulation, Section IV the proposed scheme, Section V the simulation results, and Section VI concludes the paper. Boldface lowercase and uppercase letters denote vectors and matrices; $(\cdot)^T$ and $(\cdot)^H$ denote transpose and conjugate transpose; $|\cdot|$ is the modulus of a scalar or the cardinality of a set; $\mathbb{1}\{\cdot\}$ is the indicator function; $[x]^+ = \max\{x, 0\}$; $Q(\cdot)$ is the Gaussian Q -function; and $\mathcal{CN}(0, \sigma^2)$ is the circularly symmetric complex Gaussian distribution.

II. SYSTEM MODEL

A. SWAN Architecture

We consider the downlink of a SWAN, illustrated in Fig. 1. Following [14], M dielectric waveguide segments of length L_s are placed end-to-end along the x -axis at height d above the ground plane, so that segment $m \in \mathcal{M} = \{1, \dots, M\}$ spans $[(m-1)L_s, mL_s]$ and the service area has length $D_x = ML_s$. The feed point of segment m is located at its left end, $\psi_m^0 = [(m-1)L_s, 0, d]^T$, and is connected to the BS by an individual RF chain and power amplifier with power budget $P_{\max}$. Because the segments are not physically connected, the signal captured or radiated by one segment cannot leak into another, and activating a single PA per segment avoids IAR [14]. The activated PA of segment m is located at $\psi_m = [x_m, 0, d]^T$ with

$$x_m \in \mathcal{X}_m \triangleq [(m-1)L_s + \epsilon_x, mL_s - \epsilon_x], \quad (1)$$

where ϵ_x is a small guard distance from the segment ends. We write $\mathbf{x} = [x_1, \dots, x_M]^T$.

PA reconfiguration is a physical operation. With pre-installed PAs that are switched on and off, a change of x_m takes microseconds to milliseconds; with sliding PAs it takes milliseconds to seconds [11], [23], [25]. We model a reconfiguration of segment m as a silence of τ_r slots of that segment, during which it neither transmits nor receives. Time is slotted with slot duration T_s and index $t = 0, 1, \dots$; slots are grouped into frames of $T_f \gg \tau_r$ slots. The proposed scheme changes $\mathbf{x}$ only at frame boundaries, whereas the reactive benchmark of Section V changes it within frames and pays τ_r at every change.

B. Users and Channel Model

K single-antenna HRLLC users, indexed by $k \in \mathcal{K} = \{1, \dots, K\}$, are located at $\mathbf{u}_k = [x_k, y_k, 0]^T$, uniformly distributed in $[0, D_x] \times [-D_y/2, D_y/2]$, and quasi-static within a frame; their positions are known at the BS (e.g., from the localization capability of PASS [15]). As in the PASS literature [10], [14], the link between the PA of segment m and user k is LoS dominated, and the composite channel comprises the free-space path, the in-waveguide propagation from the feed to the PA, and the in-waveguide attenuation:

$$h_{k,m}(x_m) = \frac{\sqrt{\eta} e^{-\alpha_g \ell_m}}{r_{k,m}} \exp\left(-j \frac{2\pi}{\lambda} (r_{k,m} + n_{\text{eff}} \ell_m)\right), \quad (2)$$

where $r_{k,m} = \|\mathbf{u}_k - \psi_m\| = \sqrt{(x_k - x_m)^2 + y_k^2 + d^2}$, $\ell_m = x_m - (m-1)L_s$ is the feed-to-PA distance, $\eta = c^2/(16\pi^2 f_c^2)$ with carrier frequency f_c and speed of light c , $\lambda = c/f_c$, n_{eff} is the effective refractive index of the waveguide, and $\alpha_g = \kappa \ln 10/20$ is the amplitude attenuation constant for a waveguide loss of κ dB/m. Since $\ell_m \leq L_s$, the in-waveguide loss of a SWAN is bounded by κL_s dB irrespective of D_x , which is one of the motivations for segmentation [14]. We collect the channels of user k in $\mathbf{h}_k(\mathbf{x}) = [h_{k,1}(x_1), \dots, h_{k,M}(x_M)]^T \in \mathbb{C}^M$ and define $\mathbf{H}_S(\mathbf{x}) = [\mathbf{h}_k(\mathbf{x})]_{k \in S}^H \in \mathbb{C}^{|S| \times M}$ for a user subset $S \subseteq \mathcal{K}$. To assess robustness to non-LoS scattering, the simulations optionally add a Rician component with K -factor κ_R , $h_{k,m} \leftarrow \sqrt{\frac{\kappa_R}{1+\kappa_R}} h_{k,m} + \sqrt{\frac{1}{1+\kappa_R}} |h_{k,m}| g_{k,m}$ with $g_{k,m} \sim \mathcal{CN}(0, 1)$; the design itself uses (2).

C. Transmission Modes and Finite-Blocklength Coding

All segments share the same band of B Hz, and a slot carries $n = BT_s$ channel uses. In slot t the BS selects a subset $\mathcal{S}[t] \subseteq \mathcal{K}$ of at most M users and serves them with linear precoding across the M segment feeds. Two modes arise naturally.

1) *Segment aggregation (SA):* If $|\mathcal{S}| = 1$, all segments coherently serve the single user k . Under per-segment power constraints the optimal transmission is equal-gain: every segment transmits at power $\rho P_{\max}$ with its phase aligned to the user, where $\rho \in (0, 1]$ is a power-control factor, and the received SNR is

$$\gamma_k^{\text{SA}}(\mathbf{x}, \rho) = \frac{\rho P_{\max}}{\sigma^2} \left(\sum_{m=1}^M |h_{k,m}(x_m)| \right)^2, \quad (3)$$

where σ^2 is the noise power over B Hz.

2) *Segment multiplexing (SM)*: If $2 \leq |\mathcal{S}| \leq M$, the BS applies zero-forcing (ZF) precoding $\mathbf{W}_{\mathcal{S}} = \mathbf{H}_{\mathcal{S}}^H (\mathbf{H}_{\mathcal{S}} \mathbf{H}_{\mathcal{S}}^H)^{-1}$, whose columns $\mathbf{w}_j$ satisfy $\mathbf{h}_k^H \mathbf{w}_j = \mathbb{1}\{k = j\}$, and allocates the same power p to every stream, so that segment m radiates $p \sum_{j \in \mathcal{S}} |w_{m,j}|^2$. The largest p that respects all per-segment budgets is $p(\mathcal{S}) = P_{\max} / \max_m \sum_{j \in \mathcal{S}} |w_{m,j}|^2$, and every user in $\mathcal{S}$ enjoys the interference-free SNR

$$\gamma_k^{\text{SM}}(\mathcal{S}, \mathbf{x}, \rho) = \frac{\rho p(\mathcal{S})}{\sigma^2}, \quad k \in \mathcal{S}, \quad (4)$$

with total radiated power $P_{\text{tot}}(\mathcal{S}, \rho) = \rho p(\mathcal{S}) \sum_m \sum_{j \in \mathcal{S}} |w_{m,j}|^2$. Equal-SNR ZF is chosen for its closed form; it makes the per-slot decision depend on the subset only through the two scalars $p(\mathcal{S})$ and $P_{\text{tot}}(\mathcal{S}, 1)$, which are computed once per frame for every subset (Section IV).

3) *Finite-blocklength transmission*: Packets have L bits. If user $k \in \mathcal{S}$ is served with b_k packets in slot t , they are jointly encoded into one codeword of n channel uses at rate $R_k = b_k L / n$ bits per channel use. With the normal approximation [6], the decoding error probability at SNR γ is

$$\varepsilon(b, \gamma) = Q\left(\left(\log_2(1 + \gamma) - \frac{bL}{n} + \frac{\log_2 n}{2n}\right) \sqrt{\frac{n}{V(\gamma)}}\right), \quad (5)$$

with channel dispersion $V(\gamma) = (1 - (1 + \gamma)^{-2}) \log_2^2 e$. A decoding failure is detected through the cyclic redundancy check and reported over an error-free feedback channel; the b_k packets then remain in the queue and are eligible for retransmission in the next slot. Bundling more packets into a codeword raises the rate and hence ε , so the choice of b_k is a reliability–throughput trade-off that is central to HRLLC [8], [9]; TAS resolves it per slot and per user.

D. Bursty Traffic and Deadline-Constrained Queues

Packets of user k arrive according to a Markov-modulated ON–OFF process, the standard model of bursty sources [35], [40], [41]. The source alternates between an ON state, in which packets arrive as a Poisson process with h_k packets per slot (peak rate h_k/T_s), and an OFF state without arrivals; the ON→OFF and OFF→ON transition rates are α_k and β_k (in s^{-1}), so that the mean ON and OFF durations are $1/\alpha_k$ and $1/\beta_k$ and the mean arrival rate is $\bar{\lambda}_k = h_k \beta_k / (\alpha_k + \beta_k)$ packets per slot. Users may belong to different classes: *critical* users with a larger peak rate and a tighter tail target, and *regular* users. For robustness, Section V also considers Poisson arrivals and the periodic-with-jitter model of industrial control [3] superposed with event-triggered bursts.

Each user has a first-in-first-out queue at the BS. Let $A_k[t]$ be the number of arrivals of user k in slot t and $s_k[t] \in \{0, b_k[t]\}$ the number of packets delivered (zero if user k is not scheduled or its codeword is not decoded). The delay D of a packet is the time from its arrival to its successful delivery, measured in slots and counting the transmission slot. HRLLC packets are useless after their budget: a packet whose delay would exceed $D_{\max}$ is discarded and counted as a violation, which is the deadline-dropping discipline of [5].

Denoting by $q_k[t, a]$ the number of packets of user k with age $a \in \{0, \dots, D_{\max} - 1\}$ at the beginning of slot t , the backlog is $Q_k[t] = \sum_a q_k[t, a]$ and the head-of-line (HoL) age is $\omega_k[t] = \max\{a : q_k[t, a] > 0\}$.

E. Tail-Latency Metric

The HRLLC requirement of user k is the pair $(D_{\max}, \delta_k)$: the long-run fraction of packets that are delivered later than $D_{\max}$ or discarded must not exceed δ_k ,

$$P_k^{\text{v}} \triangleq \lim_{T \rightarrow \infty} \frac{\sum_{t < T} (\text{late}_k[t] + \text{dropped}_k[t])}{\sum_{t < T} A_k[t]} = \Pr\{D_k > D_{\max}\} \leq \delta_k, \quad (6)$$

with $D_{\max} \in [0.1, 1]$ ms and $\delta_k \in [10^{-7}, 10^{-5}]$ for HRLLC [1]. Under deadline dropping, late deliveries do not occur, and P_k^{v} is the discard fraction; when a longer discard horizon is simulated to display the whole tail, both terms are counted. Note that P_k^{v} accounts for queueing, FBL decoding failures, and scheduling jointly, which is the end-to-end reliability that matters for HRLLC [42].

III. PROBLEM FORMULATION

The BS controls two sets of variables at different rates. On the frame timescale it chooses the PA positions $\mathbf{x}[f] \in \mathcal{X}_1 \times \dots \times \mathcal{X}_M$ of frame f . On the slot timescale it chooses the served subset $\mathcal{S}[t]$, the packet counts $\mathbf{b}[t] = [b_1[t], \dots, b_K[t]]$ with $b_k[t] \leq Q_k[t]$ and $b_k[t] = 0$ for $k \notin \mathcal{S}[t]$, and the power level $\rho[t]$; the precoder follows from $\mathcal{S}[t]$ and $\mathbf{x}[f]$ through (3)–(4). A policy π is the collection of these decisions. The design objective is to meet the tail targets of all users while spending as little power as possible:

$$(P1) \quad \min_{\pi} \quad \bar{P}(\pi) \triangleq \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t < T} \mathbb{E}[P_{\text{tot}}(\mathcal{S}[t], \rho[t])] \quad (7a)$$

$$\text{s.t.} \quad P_k^{\text{v}}(\pi) \leq \delta_k, \quad \forall k \in \mathcal{K}, \quad (7b)$$

$$\mathbf{x}_m[f] \in \mathcal{X}_m, \quad \forall m, f, \quad (7c)$$

$$|\mathcal{S}[t]| \leq M, \quad 0 \leq b_k[t] \leq Q_k[t], \quad \rho[t] \in (0, 1], \quad (7d)$$

$$\text{delivery and queue dynamics of Section II.} \quad (7e)$$

When the targets cannot all be met, the natural relaxation is to minimize the worst violation probability, $\min_{\pi} \max_k P_k^{\text{v}}(\pi) / \delta_k$, which is the metric reported in Section V.

Problem (P1) is a constrained Markov decision problem with a continuous geometric decision $(\mathbf{x})$, a combinatorial decision per slot (2^K subsets and $\prod_k Q_k$ packet counts), and constraints (7b) on a probability that has no closed form: the delay of a packet depends on the burst structure of all users, on the SNR tables induced by $\mathbf{x}$, on the FBL error of every bundle, and on the scheduler itself. Three observations guide the proposed solution. First, the delay tail of a single bursty source served at a constant rate is characterized in closed form by large-buffer asymptotics [32], [40], which yields a scalar rate requirement per user that captures $(h_k, \alpha_k, \beta_k, D_{\max}, \delta_k)$ jointly (Section IV-A). Second, for fixed $\mathbf{x}$ the dependence

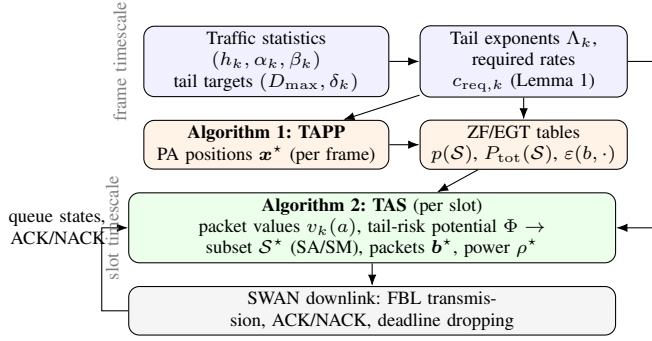


Fig. 2: Structure of TLA-SWAN. The tail exponents Λ_k enter the frame-level placement through the required rates $c_{\text{req},k}$ and the slot-level scheduler through the packet values.

of the slot-level problem on the geometry is entirely through the $O(K^M)$ scalars $\{p(\mathcal{S})\}$, which can be tabulated once per frame. Third, the constraint (7b) can be handled online by virtual queues in the drift-plus-penalty framework [48], with the objective (7a) as the penalty, which leads to a per-slot problem that is separable over subsets and users.

IV. PROPOSED TAIL-LATENCY-AWARE SCHEME: TLA-SWAN

TLA-SWAN, sketched in Fig. 2, decomposes (P1) along its two timescales. Section IV-A introduces the required-rate mapping, Section IV-B the frame-level placement (Algorithm 1), Section IV-C the slot-level scheduler (Algorithm 2), and Section IV-D the complexity and signaling. The mathematical statements are given with their proofs deferred to the appendices.

A. Tail-Latency-Aware Required Service Rate

Consider a single ON-OFF source with parameters (h, α, β) served by a constant-rate server of c packets per slot, $\bar{\lambda} < c < h$. Large-buffer asymptotics of Markov fluid queues [32], [40], [41] state that the stationary workload W satisfies $\Pr\{W > w\} \doteq e^{-\theta^*(c)w}$, where $\theta^*(c)$ is the unique positive root of $\text{EB}(\theta) = c$ and $\text{EB}(\theta)$ is the effective bandwidth of the source. Since a FIFO packet arriving to workload W experiences delay W/c , the delay-violation probability is $\Pr\{D > D_{\max}\} \doteq e^{-\theta^*(c)cD_{\max}}$. Imposing $\Pr\{D > D_{\max}\} \leq \delta$ and solving for the smallest admissible c gives the following.

Lemma 1 (Tail-latency-aware required rate). *For an ON-OFF source with peak rate h , ON $\rightarrow$ OFF rate α and OFF $\rightarrow$ ON rate β , the effective bandwidth is $\text{EB}(\theta) = \frac{1}{2\theta} [h\theta - \alpha - \beta + \sqrt{(h\theta - \alpha - \beta)^2 + 4\beta h\theta}]$, the decay rate at service rate $c \in (\bar{\lambda}, h)$ is*

$$\theta^*(c) = \frac{(\alpha + \beta)c - \beta h}{c(h - c)}, \quad (8)$$

and the smallest constant service rate for which the large-buffer approximation of $\Pr\{D > D_{\max}\}$ does not exceed δ is

$$c_{\text{req}}(h, \alpha, \beta, D_{\max}, \delta) = h \frac{\beta + \Lambda}{\alpha + \beta + \Lambda}, \quad \Lambda \triangleq \frac{\ln(1/\delta)}{D_{\max}}. \quad (9)$$

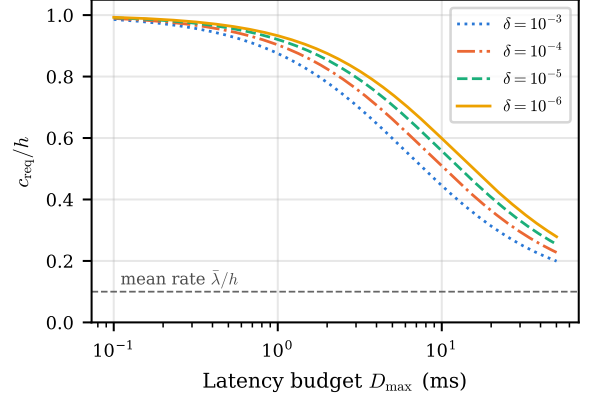


Fig. 3: Tail-latency-aware required rate (9) normalized by the peak rate h versus the latency budget $D_{\max}$, for a mean burst duration of 1 ms, an activity factor $\beta/(\alpha + \beta) = 0.1$, and several violation targets δ .

[To do] Co-author: prove Lemma 1 in Appendix A (spectral radius of the 2×2 generator plus rate matrix, the root of $\text{EB}(\theta) = c$, and the inversion of $\theta^*(c)c = \Lambda$). State precisely in which sense $\doteq$ holds (Chang's large-deviation bound, or the exact Anick-Mitra-Sondhi prefactor for a single source) and add the discrete-time/Poisson-in-ON correction if needed.

The required rate has the expected limits: $c_{\text{req}} \rightarrow \bar{\lambda}$ when $\Lambda \rightarrow 0$ (loose target), and $c_{\text{req}} \rightarrow h$ when $\Lambda \rightarrow \infty$ (stringent target); it increases with h , with the burst duration $1/\alpha$, with $\ln(1/\delta)$, and with $1/D_{\max}$. Fig. 3 illustrates (9). The scalar $c_{\text{req},k} = c_{\text{req}}(h_k, \alpha_k, \beta_k, D_{\max}, \delta_k)$ is the only traffic statistic used by the placement of Section IV-B; it can be updated online from estimates of (h_k, α_k, β_k) , and Section V shows that TLA-SWAN tolerates a mismatch of a factor of two. The tail exponent $\Lambda_k = \ln(1/\delta_k)/D_{\max}$ reappears in the scheduler of Section IV-C.

B. Frame Timescale: Tail-Aware PA Placement (TAPP)

A burst of user k is drained fastest when the SWAN aggregates all segments towards k , so the SA rate is the relevant capacity for the delay tail. Let

$$R_k^{\text{SA}}(\mathbf{x}) = \frac{n}{L} \left[\log_2(1 + \gamma_k^{\text{SA}}(\mathbf{x}, 1)) - \sqrt{\frac{V(\gamma_k^{\text{SA}})}{n}} Q^{-1}(\varepsilon_0) + \frac{\log_2 n}{2n} \right] \quad (10)$$

be the FBL rate of user k in packets per slot at the nominal reliability ε_0 [6]. We define the *tail-latency margin* of user k as $\Gamma_k(\mathbf{x}) = R_k^{\text{SA}}(\mathbf{x})/c_{\text{req},k}$: a margin below one means that even exclusive service cannot meet the tail target, while a large margin leaves room for multiplexing with other users. TAPP maximizes the worst margin,

$$(P2) \quad \max_{\mathbf{x} \in \mathcal{X}_1 \times \dots \times \mathcal{X}_M} \min_{k \in \mathcal{K}} \frac{R_k^{\text{SA}}(\mathbf{x})}{c_{\text{req},k}}. \quad (11)$$

Problem (P2) is non-convex and non-smooth, but each coordinate x_m enters γ_k^{SA} only through $|h_{k,m}(x_m)| = \sqrt{\eta} e^{-\alpha_g \ell_m} / r_{k,m}$, a smooth one-dimensional function on the interval $\mathcal{X}_m$. This structure suggests block-coordinate descent (BCD) with an exact one-dimensional search per block, as

Algorithm 1 Tail-Aware PA Placement (TAPP), executed once per frame

Require: user positions $\{u_k\}$, required rates $\{c_{\text{req},k}\}$, grid size G , tolerance ξ

- 1: Initialize $x_m \leftarrow (m - \frac{1}{2})L_s$ for all m ; $\Gamma \leftarrow \min_k \Gamma_k(x)$
- 2: **repeat**
- 3: $\Gamma_{\text{prev}} \leftarrow \Gamma$
- 4: **for** $m = 1$ **to** M **do**
- 5: Evaluate $\Gamma_k^{(g)} = \min_k \Gamma_k(x_1, \dots, x_m^{(g)}, \dots, x_M)$ on the grid $\{x_m^{(g)}\}_{g=1}^G \subset \mathcal{X}_m$
- 6: $g^* \leftarrow \arg \max_g \Gamma_k^{(g)}$; **if** $\Gamma_k^{(g^*)} > \Gamma + \xi$ **then** $x_m \leftarrow x_m^{(g^*)}$, $\Gamma \leftarrow \Gamma_k^{(g^*)}$
- 7: **end for**
- 8: **until** $\Gamma - \Gamma_{\text{prev}} \leq \xi$

Ensure: $x^* = x$; if any segment moved, silence it for τ_r slots

in Algorithm 1. The one-dimensional problem is solved by a grid of G points followed by local refinement; for a single dominant user per segment it has a closed form.

Lemma 2 (Closed-form single-user placement). *If the margin of user k is the active minimum in (P2) and only x_m is optimized, the optimal position is the projection of the user onto the segment corrected by the in-waveguide attenuation, $x_m^* = \Pi_{\mathcal{X}_m}(x_k - \zeta_m)$ with $\zeta_m \geq 0$ solving a cubic equation, and $x_m^* = \Pi_{\mathcal{X}_m}(x_k)$ when $\alpha_g = 0$. If two users k, j share the active minimum, x_m^* equalizes their margins and is the root of a quadratic equation in x_m .*

[To do] Co-author: derive Lemma 2 in Appendix B from $\partial|h_{k,m}|^2/\partial x_m = 0$ (with and without α_g) and from $\Gamma_k(x) = \Gamma_j(x)$ using the monotonicity of R^{SA} in γ ; give the explicit coefficients.

Proposition 1 (Convergence and complexity of TAPP). *The sequence of objective values generated by Algorithm 1 is non-decreasing and bounded, hence convergent, and the algorithm terminates after a finite number of sweeps for any $\xi > 0$. Each sweep costs $O(MGK)$ evaluations of (10), and the output is a coordinate-wise optimal point of (P2) on the grid.*

[To do] Co-author: prove Proposition 1 in Appendix C (monotone bounded sequence; finiteness from ξ and the boundedness of the objective by $\max_k R_k^{\text{SA}}/c_{\text{req},k}$ evaluated at $r_{k,m} = d$); discuss whether a limit point in the continuous domain is a stationary point of the max-min objective.

Remark 1. TAPP differs from rate-maximizing placement in two ways. It is max-min rather than sum, so no user is sacrificed, and the margins are normalized by $c_{\text{req},k}$, so a critical or bursty user attracts PAs even if its channel is already good. In the homogeneous case ($c_{\text{req},k} = c_{\text{req}}$ for all k) TAPP reduces to max-min SA-rate placement; the difference becomes visible with heterogeneous classes (Section V).

C. Slot Timescale: Tail-Aware Scheduling (TAS)

1) *Packet values and tail-risk potential:* Constraint (7b) is a long-term constraint on a tail probability and cannot be enforced by a per-slot rule directly. Under deadline dropping, a violation is a packet whose age reaches D_{max} , so the state

that matters for the tail is the age profile $\{q_k[t, a]\}$, and the natural Lyapunov-type quantity is one that grows as packets approach their deadlines. We assign to every queued packet of user k with age a the value

$$\nu_k(a) = \exp(\kappa \Lambda_k(a - D_{\text{max}})) = \delta_k^{\kappa(D_{\text{max}} - a)/D_{\text{max}}}, \quad \Lambda_k = \frac{\ln(1/\delta_k)}{D_{\text{max}}} \quad (12)$$

where $\kappa \in (0, 1]$ is a tail-sharpness parameter. With $\kappa = 1$ the value of a packet is its user's violation target raised to the fraction of lifetime that remains: a fresh packet is worth δ_k , a packet at its deadline is worth one, and the value multiplies by e^{Λ_k} every slot the packet waits. The exponent Λ_k is the weight of the largest-weighted-delay-first and M-LWDF rules, for which it yields the largest possible decay rate of the delay tails of all users [45], [47]; here it is applied per packet rather than per user. The *tail-risk potential* is the total value of the queued packets,

$$\Phi[t] = \sum_{k \in \mathcal{K}} \Phi_k[t], \quad \Phi_k[t] = \sum_{a=0}^{D_{\text{max}}-1} q_k[t, a] \nu_k(a), \quad (13)$$

a weighted backlog in which a packet counts more the closer it is to expiry and the tighter the target of its user. If user k delivers its b oldest packets in slot t , with ages $a_{k,(1)} \geq a_{k,(2)} \geq \dots$, the packets that would otherwise have aged into values $\nu_k(a+1)$ leave the system, and the potential decreases by

$$U_k(b) = \sum_{i=1}^b \nu_k(a_{k,(i)}), \quad \nu_k(a) \triangleq \nu_k(a+1) = \exp(\kappa \Lambda_k(a+1 - D_{\text{max}})) \quad (14)$$

Since a codeword either delivers all its packets or none, the expected decrease obtained from user $k \in \mathcal{S}$ at power level ρ is $(1 - \varepsilon(b_k, \gamma_k(\mathcal{S}, \rho)))U_k(b_k)$, with γ_k given by (3) if $|\mathcal{S}| = 1$ and by (4) otherwise. The remaining terms of the one-step drift of Φ (ageing of unserved packets, arrivals, and expiries) do not depend on the action, so that minimizing the conditional expected drift of the potential is equivalent to maximizing the expected value delivered. Adding the power penalty of the drift-plus-penalty method [48] with parameter $V \geq 0$ gives the per-slot problem

$$(P3) \max_{\mathcal{S}, \rho} \sum_{k \in \mathcal{S}} (1 - \varepsilon(b_k, \gamma_k(\mathcal{S}, \rho)))U_k(b_k) - V P_{\text{tot}}(\mathcal{S}, \rho), \quad (15)$$

subject to (7d).

2) *Solution of the per-slot problem:* For fixed $(\mathcal{S}, \rho)$, (P3) separates over users, and the optimal bundle size of user k is

$$b_k^*(\mathcal{S}, \rho) = \arg \max_{1 \leq b \leq \min\{Q_k, b_{\text{max}}\}} (1 - \varepsilon(b, \gamma_k(\mathcal{S}, \rho)))U_k(b), \quad (16)$$

a one-dimensional integer search with at most b_{max} candidates, where b_{max} is the largest number of packets that fits in a slot at the maximum SNR. Because $U_k(b)$ is increasing and concave in b while $1 - \varepsilon$ decreases in b , (16) bundles many packets when they are young and sends only the urgent ones, at a lower and more reliable rate, when the oldest packets approach their deadline; this is the tail-aware counterpart of queue-aware FBL coding [8]. The outer maximization over $(\mathcal{S}, \rho)$ is exhaustive:

Algorithm 2 Tail-Aware Scheduling (TAS), executed every slot t

Require: tables $\{p(\mathcal{S}), P_{\text{tot}}(\mathcal{S})\}_{\mathcal{S}}$, $\{\varepsilon(b, \rho p(\mathcal{S})/\sigma^2)\}_{b \leq b_{\max}, \rho \in \mathcal{R}}$ (from $\mathbf{x}^*$), age profiles $\{q_k[t, \cdot]\}$, parameters κ, V

- 1: Admit new arrivals with age 0; discard packets with age $\geq D_{\max}$ and count them as violations
- 2: Compute $v_k(a)$ by (14) and $U_k(b)$ for $b \leq \min\{Q_k, b_{\max}\}$, $\forall k$ with $Q_k > 0$
- 3: **for** every subset $\mathcal{S}$ of backlogged users with $|\mathcal{S}| \leq M$ and every $\rho \in \mathcal{R}$ **do**
- 4: $b_k^*(\mathcal{S}, \rho)$ by (16) for $k \in \mathcal{S}$; $J(\mathcal{S}, \rho)$ by (15)
- 5: **end for**
- 6: $(\mathcal{S}^*, \rho^*) \leftarrow \arg \max J(\mathcal{S}, \rho)$; transmit b_k^* packets to each $k \in \mathcal{S}^*$ with EGT if $|\mathcal{S}^*| = 1$ and ZF otherwise
- 7: On ACK, remove the b_k^* oldest packets of k ; on NACK keep them; age all remaining packets by one slot

$\gamma_k(\mathcal{S}, \rho)$ depends on $(\mathcal{S}, \rho)$ only through $\rho p(\mathcal{S})$, and the error terms $\varepsilon(b, \rho p(\mathcal{S})/\sigma^2)$ for all $b \leq b_{\max}$, all subsets, and all ρ in a finite set $\mathcal{R}$ are tabulated once per frame. With the tables in memory, evaluating (P3) for all $\sum_{s \leq M} \binom{K}{s}$ subsets is a vectorized lookup, which is feasible for $K \leq 12$ and $M \leq 8$ within a fraction of a slot (Section IV-D); for larger K the search can be restricted to the users with the largest HoL values or built greedily. The choice $|\mathcal{S}^*| = 1$ corresponds to SA and $|\mathcal{S}^*| \geq 2$ to SM, so the mode selection is a by-product of (P3) rather than a separate rule: the scheduler aggregates the segments when one user holds packets far more urgent than the others, and multiplexes when several users hold packets of comparable urgency. Algorithm 2 summarizes the procedure.

Remark 2 (Relation to virtual-queue designs). *An alternative treatment of (7b) is a virtual credit Z_k that accumulates whenever a backlogged user is served below $c_{\text{req},k}$, in the spirit of the persistent-service queues of [49], and that shifts the ages in (14) by $\zeta Z_k/c_{\text{req},k}$. We implemented this variant as well; because $c_{\text{req},k}$ is close to the peak rate for HRLLC targets, the credit grows during every burst and over-prioritizes the bursting user at the expense of the others, and Section V shows that it does not improve the tail. The packet-value formulation needs no credit and no rate estimate at the slot level.*

3) *Guarantees:* The following statements formalize the behavior of Algorithm 2.

Theorem 1 (Drift-plus-penalty characterization). *Let $\Delta(\Phi[t]) = \mathbb{E}[\Phi[t+1] - \Phi[t] | \{q_k[t, \cdot]\}]$ denote the conditional one-step drift of the tail-risk potential (13). Then (i) $\Delta(\Phi[t]) + V\mathbb{E}[P_{\text{tot}}[t] | \{q_k[t, \cdot]\}] = C[t] - \sum_{k \in \mathcal{S}[t]} e^{\kappa \Lambda_k} (1 - \varepsilon_k[t]) U_k(b_k[t]) + V P_{\text{tot}}[t]$, where $C[t]$ does not depend on the action, so that Algorithm 2 with the user weights $e^{\kappa \Lambda_k}$ absorbed in $v_k(\cdot)$ minimizes the drift-plus-penalty expression exactly in every slot; (ii) if the arrivals are bounded, $A_k[t] \leq A_{\max}$, and there exists a stationary randomized policy π and $\epsilon > 0$ such that $\mathbb{E}_\pi[\text{value delivered in a slot} | \Phi] \geq (e^{\kappa \Lambda_{\max}} - 1)\Phi / e^{\kappa \Lambda_{\max}} + \epsilon \Phi$ whenever Φ exceeds a*

threshold, then the time-average potential under Algorithm 2 with $V = 0$ is bounded, $\bar{\Phi} \leq \mathcal{B}_\Phi/\epsilon$, and the delay-violation probability of user k satisfies $P_k^v \leq e^{\kappa \Lambda_k} \bar{\Phi}_k / \bar{\lambda}_k$, where $\bar{\Phi}_k$ is the time-average of Φ_k ; (iii) for $V > 0$ the time-average power is within $O(1/V)$ of that of π at the price of an $O(V)$ increase of $\bar{\Phi}$.

[To do] Co-author: prove Theorem 1 in Appendix D. Part (i) is direct from the dynamics of Φ : unserved packets multiply their value by $e^{\kappa \Lambda_k}$, arrivals add $\nu_k(0) = \delta_k^\kappa$ each, expiring packets remove $\nu_k(D_{\max}-1)$ each, and delivered packets remove $U_k(b_k)$. For (ii), use that every discarded packet had value $\nu_k(D_{\max}-1) = e^{-\kappa \Lambda_k}$ in the slot before its discard, so the discard rate is bounded by $e^{\kappa \Lambda_k}$ times the rate at which value leaves Φ_k through expiry, which is at most the time-average increase of Φ_k from ageing; combine with the telescoping sum of the drift and the per-slot optimality of (i). Adapt the drift-plus-penalty theorem of [48, Ch. 4] to the multiplicative (rather than additive) dynamics of Φ ; state the constants $\mathcal{B}_\Phi$ and the threshold explicitly, and check whether the slack condition can be expressed through $c_{\text{req},k}$ and the SA rates $R_k^{\text{SA}}(\mathbf{x}^*)$ of Lemma 1 and Section IV-B.

Theorem 2 (Tail bound). *Let $\mu_k(\theta)$ denote the effective capacity [34], [43] of the service process $\{s_k[t]\}$ of user k induced by TLA-SWAN, and let $\text{EB}_k(\theta)$ be the effective bandwidth of its ON-OFF source. If $\theta_k^* > 0$ solves $\text{EB}_k(\theta) = \mu_k(\theta)$, then $\Pr\{D_k > D_{\max}\} \leq \varsigma_k e^{-\theta_k^* \text{EB}_k(\theta_k^*) D_{\max}}$ with $\varsigma_k = \Pr\{Q_k > 0\} \leq 1$. In particular, if the fraction of slots in which user k is served in SA mode is at least ϕ_k and the SA bundle is decoded with probability at least $1 - \epsilon_0$, then $\mu_k(\theta) \geq -\frac{1}{\theta} \ln(1 - \phi_k + \phi_k(\epsilon_0 + (1 - \epsilon_0)e^{-\theta b_k^{\text{SA}}}))$, where $b_k^{\text{SA}} = \lfloor R_k^{\text{SA}}(\mathbf{x}^*) \rfloor$, which makes the bound computable from the output of Algorithm 1.*

[To do] Co-author: prove Theorem 2 in Appendix E via the effective bandwidth/effective capacity duality [32], [34] with the FBL service model of [38], [43] (i.i.d. success indicators conditioned on the schedule), and justify the lower bound on $\mu_k(\theta)$ by the monotonicity of the effective capacity in the service distribution. Discuss the choice of ϕ_k (e.g., $\phi_k = \min\{1, c_{\text{req},k}/b_k^{\text{SA}}\}$) and compare the bound with the simulated tail in Fig. ?? (to be added once derived).

D. Complexity and Signaling

TAPP runs once per frame with $O(IMGK)$ operations, where I is the number of sweeps ($I \leq 5$ in all our experiments). The per-frame tables require one ZF inversion of size $|\mathcal{S}| \times |\mathcal{S}|$ per subset, i.e., $O(\sum_{s \leq M} \binom{K}{s} s^3)$, and the FBL error table has $|\mathcal{R}| b_{\max}$ entries per subset; for $K = 8$, $M = 4$, $|\mathcal{R}| = 5$, $b_{\max} = 12$ this is 162 subsets and about 10^4 scalars. Per slot, TAS evaluates the values (14) in $O(KD_{\max})$ and (P3) in $O(|\mathcal{R}| b_{\max} \sum_{s \leq M} \binom{K}{s})$ vectorized operations, which is a lookup of the order of 10^4 scalars, i.e., microseconds on a modern processor. Signaling is that of any multi-user MIMO downlink (CSI at the BS, ACK/NACK feedback) plus the traffic statistics of each user, which are three scalars updated on the frame timescale; the PA positions are set by the BS itself.

TABLE I: Default simulation parameters

Parameter	Value	Parameter	Value
Carrier f_c	28 GHz	Segments M , length L_s	4, 10 m
n_{eff}, κ	1.4, 0.08 dB/m	Height d , width D_y	5 m, 10 m
Noise σ^2 ($B = 2$ MHz)	-101 dBm	Users K	8
Per-segment power P_{max}	-10 dBm	Slot T_s , blocklength n	0.1 ms, 200
Packet L	256 bits	$b_{\text{max}}, \mathcal{R}$	12, $\{1\}$ ($\{0.1, \dots, 1\}$ for $V > 0$)
Peak rate h	3 pkt/slot	ON/OFF means $1/\alpha, 1/\beta$	1, 5
Budget D_{max}	1 ms (10 slots)	Target δ	10^{-5}
Critical users	$h = 4, \delta = 10^{-6}$	Nominal ε_0 (TAPP)	10^{-5}
Reconfiguration τ_r	1 slot	TAS κ, V	0.75, 0

V. SIMULATION RESULTS

A. Setup and Benchmarks

We simulate the system of Section II at the slot level, including the FBL decoding errors drawn from (5), retransmissions, deadline dropping, the reconfiguration silence, and the exact packet ages, with the parameters of Table I unless stated otherwise. The PHY parameters follow the SWAN literature [14], [17]; the slot of 0.1 ms with $n = 200$ channel uses corresponds to a 2-MHz HRLLC carrier and the 32-byte packets to the 3GPP reference payload [3], [57]. The default traffic is an ON-OFF source with mean burst duration $1/\alpha = 1$ ms, activity factor $\beta/(\alpha + \beta) = 0.1$, and peak rate $h = 3$ packets per slot, i.e., a peak of 7.7 Mb/s and a mean load of 0.3 packets per slot per user; for $K = 8$ users the mean offered load is 2.4 packets per slot, whereas a single stream carries between 3 and 5 packets per slot depending on the SNR, so that a single burst is drainable but simultaneous bursts overload the system. Each point pools four independent user drops of 2.5×10^5 slots (25 s) each, i.e., about 2.4×10^6 packets per point, so that violation probabilities down to 10^{-5} are resolved; the delay tail of Fig. ?? uses 3.2×10^6 slots. Points with zero observed violations are drawn at $1/(\text{number of packets})$.

The following schemes are compared.

- **TLA-SWAN**: TAPP placement and TAS scheduling (proposed).
- **SWAN, M-LWDF**: TAPP placement with the M-LWDF rule [45], i.e., user weights $a_k \omega_k[t]/\bar{r}_k[t]$ with $\bar{r}_k$ the exponentially averaged delivered rate, applied to the same subset search and goodput-maximizing bundling.
- **SWAN, max-weight**: TAPP placement with queue-length weights (throughput-optimal, tail-agnostic).
- **SWAN, EDF**: TAPP placement with earliest-deadline-first user selection, i.e., the M users with the oldest HoL packets, without channel awareness.
- **SWAN, sum-rate + MW**: PA placement maximizing $\sum_k R_k^{\text{SA}}$ (the rate-centric SWAN design [14], [16]) with max-weight scheduling.
- **SWAN, fixed PAs**: PAs at the segment centers (no reconfiguration) with TAS.
- **SWAN, reactive**: every slot, each segment moves its PA above the backlogged user with the largest weight in its range and pays τ_r slots of silence per move; TAS scheduling.
- **Single-waveguide PASS**: one waveguide of length D_x fed at $x = 0$ with one RF chain, M PAs at the TAPP positions,

in-waveguide loss over the full length, and SA/TDMA only (one user per slot) [10].

- **Fixed antenna array**: M co-located antennas at $\lambda/2$ spacing at the center of the area at height d , ZF precoding, TAS.

VI. CONCLUSIONS

This paper has studied, for the first time, a segmented waveguide-enabled pinching-antenna system that serves HRLLC users with bursty, deadline-constrained traffic and finite blocklength codes. The delay tail, not the rate, was taken as the design target, and it was shown that this changes both where the pinching antennas should be placed and how the segments should be used. The tail-latency-aware required rate summarizes the burst statistics and the tail target of a user in one closed-form scalar; maximizing the minimum ratio between the aggregation-mode rate and this scalar moves the antennas towards the users whose bursts are hardest to drain, and a packet-value scheduler that values every packet by its remaining lifetime and tail target, and minimizes the drift of the resulting tail-risk potential, selects, slot by slot, between segment aggregation and segment multiplexing, the number of packets per codeword, and the power level. The slot-level simulations, which include the decoding errors, retransmissions, deadline dropping, and reconfiguration delays that determine the tail, showed order-of-magnitude reductions of the delay-violation probability with respect to rate-centric SWAN designs, classical deadline-aware schedulers, single-waveguide PASS, and fixed arrays, robustness to fading, traffic models, and parameter mismatch, and the superiority of statistical placement over reactive repositioning whenever reconfiguration is not free. Future work includes the joint design with segment activation under a limited number of RF chains, uplink and grant-free access with tail guarantees, learning the traffic statistics online, and the extension of the analysis to correlated bursts across users.

APPENDIX A PROOF OF LEMMA 1

[To do] Co-author: derivation of the effective bandwidth, of (8), and of (9); see docs/math_todo_for_colleague.md, item L1.

APPENDIX B PROOF OF LEMMA 2

[To do] Co-author: item L2.

APPENDIX C PROOF OF PROPOSITION 1

[To do] Co-author: item P1.

APPENDIX D PROOF OF THEOREM 1

[To do] Co-author: item T1.

APPENDIX E PROOF OF THEOREM 2

[To do] Co-author: item T2.

REFERENCES

- [1] ITU-R, "Framework and overall objectives of the future development of IMT for 2030 and beyond," International Telecommunication Union, Tech. Rep. Recommendation ITU-R M.2160-0, Nov. 2023.
- [2] M. Bennis, M. Debbah, and H. V. Poor, "Ultrareliable and low-latency wireless communication: Tail, risk, and scale," *Proc. IEEE*, vol. 106, no. 10, pp. 1834–1853, Oct. 2018.
- [3] 3GPP, "Service requirements for cyber-physical control applications in vertical domains (Release 17)," 3rd Generation Partnership Project, Tech. Rep. TS 22.104 V17.7.0, May 2022.
- [4] —, "Study on XR (extended reality) evaluations for NR (Release 17)," 3rd Generation Partnership Project, Tech. Rep. TR 38.838 V17.0.0, Dec. 2021.
- [5] L. Zhang, A. Liu, and K. Wang, "A hybrid reinforcement learning framework for hard latency constrained resource scheduling," *arXiv preprint arXiv:2504.03721*, 2025.
- [6] Y. Polyanskiy, H. V. Poor, and S. Verdú, "Channel coding rate in the finite blocklength regime," *IEEE Trans. Inf. Theory*, vol. 56, no. 5, pp. 2307–2359, May 2010.
- [7] G. Durisi, T. Koch, and P. Popovski, "Toward massive, ultrareliable, and low-latency wireless communication with short packets," *Proc. IEEE*, vol. 104, no. 9, pp. 1711–1726, Sep. 2016.
- [8] X. Zhao, W. Chen, and H. V. Poor, "Queue-aware finite-blocklength coding for ultra-reliable and low-latency communications: A cross-layer approach," *IEEE Trans. Wireless Commun.*, vol. 21, no. 10, pp. 8786–8802, Oct. 2022.
- [9] L. Li, W. Chen, P. Popovski, and K. B. Letaief, "Reliability-latency-rate tradeoff in low-latency communications with finite-blocklength coding," *IEEE Trans. Inf. Theory*, 2024.
- [10] Z. Ding, R. Schober, and H. V. Poor, "Flexible-antenna systems: A pinching-antenna perspective," *IEEE Trans. Commun.*, vol. 73, no. 6, pp. 4232–4247, Jun. 2025.
- [11] Y. Liu, H. Jiang, X. Xu, Z. Wang, J. Guo, C. Ouyang, X. Mu, Z. Ding, A. Nallanathan, G. K. Karagiannidis, and R. Schober, "Pinching-antenna systems (PASS): A tutorial," *IEEE Trans. Commun.*, vol. 74, pp. 4881–4918, 2026.
- [12] Y. Liu, H. Jiang, X. Gan, X. Xu, J. Guo, Z. Wang, C. Ouyang, X. Mu, Z. Ding, A. Nallanathan, O. A. Dobre, G. K. Karagiannidis, and R. Schober, "A survey of pinching-antenna systems (PASS)," *IEEE Commun. Surveys Tuts.*, 2026, to appear, arXiv:2601.18927.
- [13] Y. Xu, J. Cui, Y. Zhu, Z. Ding, T.-H. Chang, R. Schober, V. W. S. Wong, O. A. Dobre, G. K. Karagiannidis, H. V. Poor, and X. You, "Generalized pinching-antenna systems: A tutorial on principles, design strategies, and future directions," *arXiv preprint arXiv:2510.14166*, 2025.
- [14] C. Ouyang, H. Jiang, Z. Wang, Y. Liu, and Z. Ding, "Uplink and downlink communications in segmented waveguide-enabled pinching-antenna systems (SWANs)," *IEEE Trans. Commun.*, vol. 74, pp. 3688–3703, 2026.
- [15] S. Shan, C. Ouyang, P. Popovski, and Y. Liu, "Access protocols for segmented waveguide-enabled pinching-antenna systems (SWANs)," *arXiv preprint arXiv:2606.04913*, 2026.
- [16] S. Gu, H. Jiang, C. Ouyang, Y. Liu, and D. I. Kim, "Sum-rate maximization for uplink segmented waveguide-enabled pinching-antenna systems," *IEEE Wireless Commun. Lett.*, vol. 15, pp. 3876–3880, 2026.
- [17] S. Gu, Z. Cheng, H. Jiang, C. Ouyang, Y. Liu, and A. Nallanathan, "Joint segment activation and antenna placement for uplink SWAN systems," *arXiv preprint arXiv:2605.13580*, 2026.
- [18] H. Jiang, C. Ouyang, Z. Wang, Y. Liu, A. Nallanathan, Z. Ding, and R. Schober, "Exploiting segmented waveguide-enabled pinching-antenna systems (SWANs) for uplink tri-hybrid beamforming," *arXiv preprint arXiv:2603.04684*, 2026.
- [19] J. Guo, C. Ouyang, Y. Liu, A. Nallanathan, and Z. Ding, "Uplink segmented waveguide-enabled multiuser pinching-antenna systems: Protocol-aware graph neural network-based antenna placement," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 22 227–22 244, 2026.
- [20] H. Jiang, C. Ouyang, Z. Wang, Y. Liu, A. Nallanathan, Z. Ding, and R. Schober, "Segmented waveguide-enabled pinching-antenna systems (SWANs) for ISAC," *arXiv preprint arXiv:2512.07649*, 2025.
- [21] Q. Gao, R. Zhong, H. Shin, and Y. Liu, "Integrated sensing and communication for segmented waveguide-enabled pinching antenna systems," *arXiv preprint arXiv:2601.20658*, 2026.
- [22] Y. Geng, T. H. Cheng, K. C. Teh, and Z. Ding, "CRLB minimization for ISAC systems with segmented waveguide-enabled pinching antenna," *arXiv preprint arXiv:2604.00572*, 2026.
- [23] K. Wang, D. K. C. So, and Z. Ding, "Pinching-antenna systems: From antenna placement to antenna roaming," *arXiv preprint arXiv:2608.10136*, 2026.
- [24] —, "Age-of-information aware federated learning with finite speed pinching antenna," *arXiv preprint arXiv:2607.23595*, 2026.
- [25] J. Jiang, X. Xu, C.-T. Lam, Y. Liu, and A. Nallanathan, "Discrete coupling and localized motion for pinching-antenna systems (PASS)," *arXiv preprint arXiv:2609.16293*, 2026.
- [26] C. Ouyang, H. Jiang, Z. Wang, Y. Liu, and Z. Ding, "On the maintainability of pinching-antenna systems: A failure-repair perspective," *IEEE Trans. Wireless Commun.*, 2026, to appear, arXiv:2602.11784.
- [27] C. Ouyang, H. Jiang, Z. Ding, and R. Schober, "Frequency-selective pinching-antenna systems," *arXiv preprint arXiv:2608.21020*, 2026.
- [28] Y. Zheng, S. L. Cotton, and T. Q. Duong, "A bidirectional-AoI-aware multi-agent deep reinforcement learning framework for vehicular platooning in segmented waveguide-based pinching antenna systems," *IEEE Trans. Cogn. Commun. Netw.*, vol. 12, pp. 10965–10978, 2026.
- [29] T. Lin, J. Zhu, W. Huang, M. Hua, and Z. Zhang, "Rate optimization for downlink URLLC via pinching antenna arrays," *arXiv preprint arXiv:2509.01222*, 2025.
- [30] Z. Pan, G. Zheng, Z. Xu, and L. Yuan, "Statistical CSI-based design for pinching antenna systems with short-packet communication," *Entropy*, vol. 28, no. 7, p. 722, 2026.
- [31] Y. Xu, S. Shan, Y. Zhu, Z. Ding, M. Peng, and X. You, "Pinching antennas for next-generation wireless communications: A network perspective," *arXiv preprint arXiv:2609.17332*, 2026.
- [32] C.-S. Chang, "Stability, queue length, and delay of deterministic and stochastic queueing networks," *IEEE Trans. Autom. Control*, vol. 39, no. 5, pp. 913–931, May 1994.
- [33] F. P. Kelly, "Notes on effective bandwidths," in *Stochastic Networks: Theory and Applications*, F. P. Kelly, S. Zachary, and I. Ziedins, Eds. Oxford Univ. Press, 1996, pp. 141–168.
- [34] D. Wu and R. Negi, "Effective capacity: A wireless link model for support of quality of service," *IEEE Trans. Wireless Commun.*, vol. 2, no. 4, pp. 630–643, Jul. 2003.
- [35] M. Ozmen and M. C. Gursoy, "Wireless throughput and energy efficiency with random arrivals and statistical queuing constraints," *IEEE Trans. Inf. Theory*, vol. 62, no. 3, pp. 1375–1395, Mar. 2016.
- [36] M. Fidler and A. Rizk, "A guide to the stochastic network calculus," *IEEE Commun. Surveys Tuts.*, vol. 17, no. 1, pp. 92–105, 2015.
- [37] H. Al-Zubaidy, J. Liebeherr, and A. Burchard, "Network-layer performance analysis of multihop fading channels," *IEEE/ACM Trans. Netw.*, vol. 24, no. 1, pp. 204–217, Feb. 2016.
- [38] S. Schiessl, H. Al-Zubaidy, M. Skoglund, and J. Gross, "Delay performance of wireless communications with imperfect CSI and finite-length coding," *IEEE Trans. Commun.*, vol. 66, no. 12, pp. 6527–6541, Dec. 2018.
- [39] C.-F. Liu, M. Bennis, M. Debbah, and H. V. Poor, "Dynamic task offloading and resource allocation for ultra-reliable low-latency edge computing," *IEEE Trans. Commun.*, vol. 67, no. 6, pp. 4132–4150, Jun. 2019.
- [40] D. Anick, D. Mitra, and M. M. Sondhi, "Stochastic theory of a data-handling system with multiple sources," *Bell Syst. Tech. J.*, vol. 61, no. 8, pp. 1871–1894, Oct. 1982.
- [41] A. I. Elwalid and D. Mitra, "Effective bandwidth of general Markovian traffic sources and admission control of high speed networks," *IEEE/ACM Trans. Netw.*, vol. 1, no. 3, pp. 329–343, Jun. 1993.
- [42] C. She, C. Yang, and T. Q. S. Quek, "Cross-layer optimization for ultra-reliable and low-latency radio access networks," *IEEE Trans. Wireless Commun.*, vol. 17, no. 1, pp. 127–141, Jan. 2018.
- [43] M. C. Gursoy, "Throughput analysis of buffer-constrained wireless systems in the finite blocklength regime," *EURASIP J. Wireless Commun. Netw.*, vol. 2013, no. 290, 2013.
- [44] L. Li and W. Chen, "A tractable approach for queueing analysis on buffer-aware scheduling," *arXiv preprint arXiv:2412.18812*, 2024.
- [45] M. Andrews, K. Kumaran, K. Ramanan, A. Stolyar, P. Whiting, and R. Vijayakumar, "Providing quality of service over a shared wireless link," *IEEE Commun. Mag.*, vol. 39, no. 2, pp. 150–154, Feb. 2001.
- [46] S. Shakkottai and A. L. Stolyar, "Scheduling for multiple flows sharing a time-varying channel: The exponential rule," in *Analytic Methods in Applied Probability: In Memory of Fridrikh Karpelevich*, ser. American Mathematical Society Translations, Series 2. Amer. Math. Soc., 2002, vol. 207, pp. 185–202.
- [47] A. L. Stolyar and K. Ramanan, "Largest weighted delay first scheduling: Large deviations and optimality," *Ann. Appl. Probab.*, vol. 11, no. 1, pp. 1–48, 2001.

- [48] M. J. Neely, *Stochastic Network Optimization with Application to Communication and Queueing Systems*. Morgan & Claypool, 2010.
- [49] —, “Delay-based network utility maximization,” *IEEE/ACM Trans. Netw.*, vol. 21, no. 1, pp. 41–54, Feb. 2013.
- [50] C. Zhang, L. Wei, J. Fan, Z. Liu, and Y. Huang, “Lyapunov-guided multi-agent reinforcement learning for delay-sensitive wireless scheduling,” *arXiv preprint arXiv:2411.01766*, 2024.
- [51] K. Zambouri, M. Noor-A-Rahim, C. J. Sreenan, and D. Pesch, “Overload-robust latency in 5G-TSN: A HoL-enhanced hybrid Lyapunov approach for 3GPP indoor factory environments,” *arXiv preprint arXiv:2609.26011*, 2026.
- [52] X. Zhu, K.-K. Wong, H. Xu, C. Rao, and H. Shin, “Fluid antenna systems enabling 6G HRLLC with port switching delay,” *arXiv preprint arXiv:2605.06275*, 2026.
- [53] Z. Zhang, K.-K. Wong, D. Morales-Jimenez, H. Jiang, H. Xu, C. Masouros, Z. Zhang, and C.-B. Chae, “Finite-blocklength fluid antenna systems,” *arXiv preprint arXiv:2509.15643*, 2025.
- [54] X. He, W. Chen, Q. Wu, X. Zhu, and N. Cheng, “Movable antenna enhanced NOMA short-packet transmission,” *IEEE Commun. Lett.*, vol. 28, no. 9, pp. 2196–2200, Sep. 2024.
- [55] R. Hashemi, S. Ali, N. H. Mahmood, and M. Latva-aho, “Average rate and error probability analysis in short packet communications over RIS-aided URLLC systems,” *IEEE Trans. Veh. Technol.*, vol. 70, no. 10, pp. 10 320–10 334, Oct. 2021.
- [56] Q. Peng *et al.*, “Two-timescale design for reconfigurable intelligent surface-aided URLLC,” *IEEE Trans. Wireless Commun.*, 2024.
- [57] 3GPP, “Service requirements for the 5G system,” 3rd Generation Partnership Project, Tech. Rep. TS 22.261, 2023.